

APSPDCL · COMMERCIAL ROOFTOP

350 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.35 Cr after input tax credit, generating 5,36,550 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹2.49 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	350 kWp
Year 1 generation	5,36,550 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹38.42 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹1.42 Cr
GST at 5% — recovered as ITC	₹6.75 L
Net capitalised cost	₹1.35 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹33.98 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,35,00,000	54,00,000	81,00,000	13,59,072
2	81,00,000	32,40,000	48,60,000	8,15,443
3	48,60,000	19,44,000	29,16,000	4,89,266
4	29,16,000	11,66,400	17,49,600	2,93,560
5	17,49,600	6,99,840	10,49,760	1,76,136
6	10,49,760	4,19,904	6,29,856	1,05,681
7	6,29,856	2,51,942	3,77,914	63,409
8	3,77,913	1,51,165	2,26,748	38,045
9	2,26,748	90,699	1,36,049	22,827
10	1,36,049	54,420	81,629	13,696
11	81,630	32,652	48,978	8,218
12	48,978	19,591	29,387	4,931
Tax shield, years 1-12				33,90,284

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

350 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	5,36,550	38,41,698	—	54,00,000	(3,92,193)	42,33,891	42,33,891
2	5,33,867	39,75,389	2,10,000	32,40,000	1,32,230	36,33,159	78,67,050
3	5,31,198	41,13,733	2,10,000	19,44,000	4,93,226	34,10,507	1,12,77,557
4	5,28,542	42,56,891	2,10,000	11,66,400	7,24,962	33,21,929	1,45,99,486
5	5,25,899	44,05,030	2,10,000	6,99,840	8,79,669	33,15,361	1,79,14,847
6	5,23,270	45,58,325	2,10,000	4,19,904	9,88,705	33,59,620	2,12,74,467
7	5,20,653	47,16,955	2,10,000	2,51,942	10,70,902	34,36,053	2,47,10,520
8	5,18,050	48,81,105	2,10,000	1,51,165	11,37,578	35,33,527	2,82,44,047
9	5,15,460	50,50,968	2,10,000	90,699	11,95,548	36,45,420	3,18,89,467
10	5,12,883	52,26,741	2,10,000	54,420	12,48,917	37,67,824	3,56,57,291
11	5,10,318	54,08,632	2,10,000	32,652	13,00,174	38,98,458	3,95,55,749
12	5,07,767	55,96,852	2,10,000	19,591	13,50,832	40,36,020	4,35,91,769
13	5,05,228	57,91,623	2,10,000	11,755	14,01,824	41,79,798	4,77,71,567
14	5,02,702	59,93,171	2,10,000	7,053	14,53,733	43,29,438	5,21,01,005
15	5,00,188	62,01,734	2,10,000	4,232	15,06,934	44,84,799	5,65,85,804
16	4,97,687	64,17,554	2,10,000	2,539	15,61,678	46,45,876	6,12,31,680
17	4,95,199	66,40,885	2,10,000	1,523	16,18,142	48,12,743	6,60,44,423
18	4,92,723	68,71,988	2,10,000	914	16,76,459	49,85,529	7,10,29,952
19	4,90,259	71,11,133	2,10,000	548	17,36,739	51,64,394	7,61,94,346
20	4,87,808	73,58,600	2,10,000	329	17,99,077	53,49,523	8,15,43,869
21	4,85,369	76,14,679	2,10,000	197	18,63,560	55,41,119	8,70,84,988
22	4,82,942	78,79,670	2,10,000	118	19,30,273	57,39,397	9,28,24,385
23	4,80,527	81,53,883	2,10,000	71	19,99,299	59,44,584	9,87,68,969
24	4,78,125	84,37,638	2,10,000	43	20,70,721	61,56,917	10,49,25,886
25	4,75,734	87,31,268	2,10,000	26	21,44,626	63,76,641	11,13,02,527
26	4,73,355	90,35,116	2,10,000	15	22,21,101	66,04,015	11,79,06,542
27	4,70,988	93,49,538	2,10,000	9	23,00,237	68,39,301	12,47,45,843
28	4,68,634	96,74,902	2,10,000	6	23,82,125	70,82,777	13,18,28,620
29	4,66,290	1,00,11,588	2,10,000	3	24,66,863	73,34,725	13,91,63,345
30	4,63,959	1,03,59,992	2,10,000	2	25,54,549	75,95,442	14,67,58,787
Total	1,49,82,174	19,76,67,281	60,90,000	1,34,99,996	4,48,18,490	14,67,58,787	14,67,58,787

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹2.49 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹14.68 Cr
30-year generation	1,49,82,172 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,49,82,172 units, avoiding 12,285.4 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.42 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 350 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.